

# Homework: DreamBooth Fine-Tuning Practice

## Overview

In this project you will run a small DreamBooth-style fine-tuning experiment for subject-driven image generation. The goal is to go through the full loop: **collect data** → **fine-tune a model** → **generate images** → **analyze the results**

The emphasis is not on achieving perfect photorealism, but on understanding how a generative model learns a **specific visual identity** and how training choices affect the outcome.

This assignment is inspired by the DreamBooth paper.

## Choosing a Subject

Each student should work with a **unique subject**.

Typical choices include things like:

- a pet
- a specific toy
- a personal object
- a figurine or plush

Avoid generic internet images that other students might easily use. The subject should be something **distinct and identifiable**.

You will collect **3–5 images of the same subject** to use for training.

## Model Choice

Any **open-source diffusion model** that can support DreamBooth-style fine-tuning is acceptable.

Common choices include:

- Stable Diffusion v1.5
- Stable Diffusion 2.x
- Stable Diffusion XL
- other comparable open models

Stable Diffusion 1.5 is usually the easiest starting point due to its size and tooling support.

Parameter-efficient approaches such as **LoRA-based DreamBooth** are also acceptable.

## Preparing the Data

DreamBooth typically associates a **unique identifier token** with a subject.

For example:

**“a photo of sks dog”**

Here:

- **sks** is a rare token acting as the identifier
- **dog** represents the class category

During training the model learns to associate the identifier with the specific subject in your images.

When preparing the images, it is generally helpful to ensure that:

- the subject is clearly visible
- images have reasonable resolution
- different images show slightly different angles or contexts

Exact preprocessing details may depend on the model and training pipeline you use.

## Training

Use any DreamBooth implementation you prefer. Examples include:

- Hugging Face diffusers DreamBooth scripts
- kohya\_ss
- other reproducible implementations

You are free to decide how to run the experiment. For example, you may choose:

- full model fine-tuning
- LoRA-based fine-tuning
- freezing or training the text encoder
- different training lengths or learning rates

In your report, briefly document the main training configuration so that the experiment can be reproduced.

## Generating Images

After training, generate images using prompts that include your identifier token.

It is often interesting to try prompts that place the subject in **new contexts or styles**, such as:

- different environments
- modified attributes (color, material, etc.)
- added accessories
- artistic styles

These examples help reveal how well the model has learned the subject and how flexible the generation remains.

You are free to design your own prompt set.

## Evaluation and Discussion

When looking at the generated images, two questions are often useful:

- **Subject fidelity:** Does the generated image resemble the specific subject from the training images?
- **Prompt fidelity:** Does the image follow the prompt (environment, style, objects, etc.)?

DreamBooth training often involves a trade-off between these two properties.

Typical behaviors include:

- **Overfitting:** the subject looks correct but the model ignores new prompts or reproduces the original training background.
- **Underfitting:** the prompt is followed but the subject identity is weak or inconsistent.

Discuss briefly what you observe in your results.

## Optional: Diversity and Prior Preservation

The DreamBooth paper also discusses two additional aspects that can be useful when analyzing results:

### Diversity

A model that overfits heavily may produce images that look very similar to each other. You may optionally examine how diverse your generated images are across different prompts or random seeds. In research papers, diversity is sometimes measured using metrics such as **LPIPS** between generated samples, but a qualitative comparison is also acceptable.

### Prior preservation

DreamBooth training may cause the model to collapse toward the instance subject and forget the broader class distribution (e.g., generating your specific dog even when prompted with just “a dog”). You may optionally test this by generating images using the **class prompt only** and observing whether the model still produces diverse class examples rather than copies of the trained subject.

These analyses are optional but can provide additional insight into how the fine-tuning process affects the model.

## Project Outputs

Your submission should include:

1. **A small dataset**  
3–5 images of a specific subject.

2. **Training code or notebook**

A Colab notebook or script that performs the fine-tuning and generation process.

3. **Generated images**

Examples showing how the trained model renders the subject.

4. **A short report (PDF)** describing

- the subject and dataset
- the model and training setup
- representative results
- observations about model behavior

The report does not need to be long. Clear figures and concise discussion are sufficient.

## Submission

Please submit:

1. A zipped folder containing your training images.
2. The notebook or scripts used for training and generation.
3. A short **PDF report** including
  - representative generated images
  - key training settings
  - brief analysis of the results.